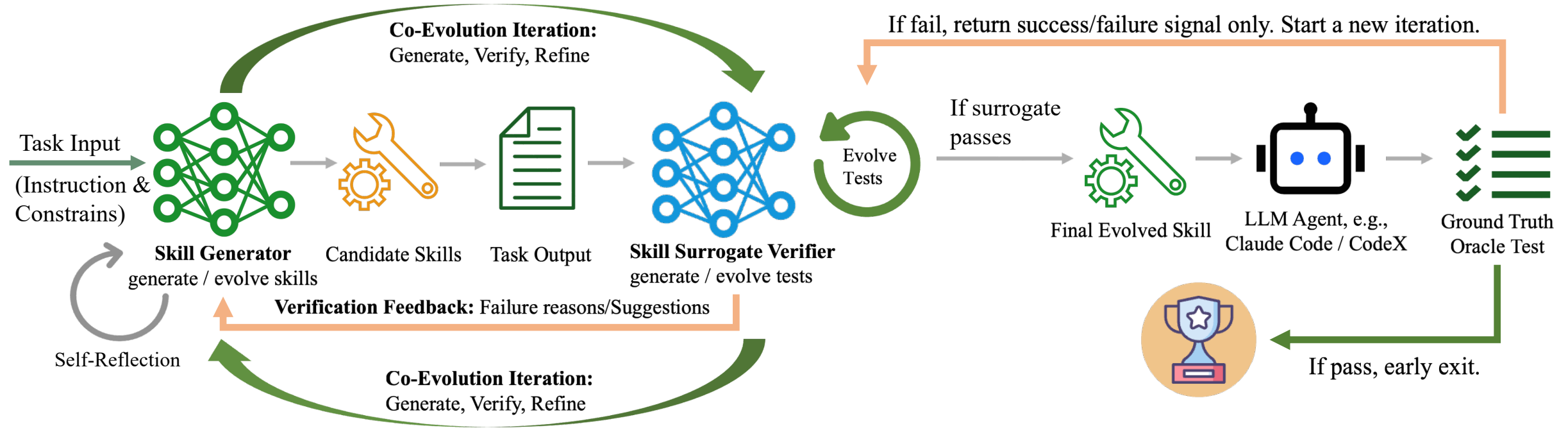
Co-Evolution Iteration:
Generate, Verify, Refine
If fail, return success/failure signal only. Start a new iteration.
Task Input
(Instruction & Constrains)
Skill Generator
generate / evolve skills
Candidate Skills
Task Output
Skill Surrogate Verifier
generate / evolve tests
Evolve Tests
If surrogate passes
Final Evolved Skill
LLM Agent, e.g., Claude Code / CodeX
Ground Truth Oracle Test
Verification Feedback: Failure reasons/Suggestions
Self-Reflection
Co-Evolution Iteration:
Generate, Verify, Refine
If pass, early exit.